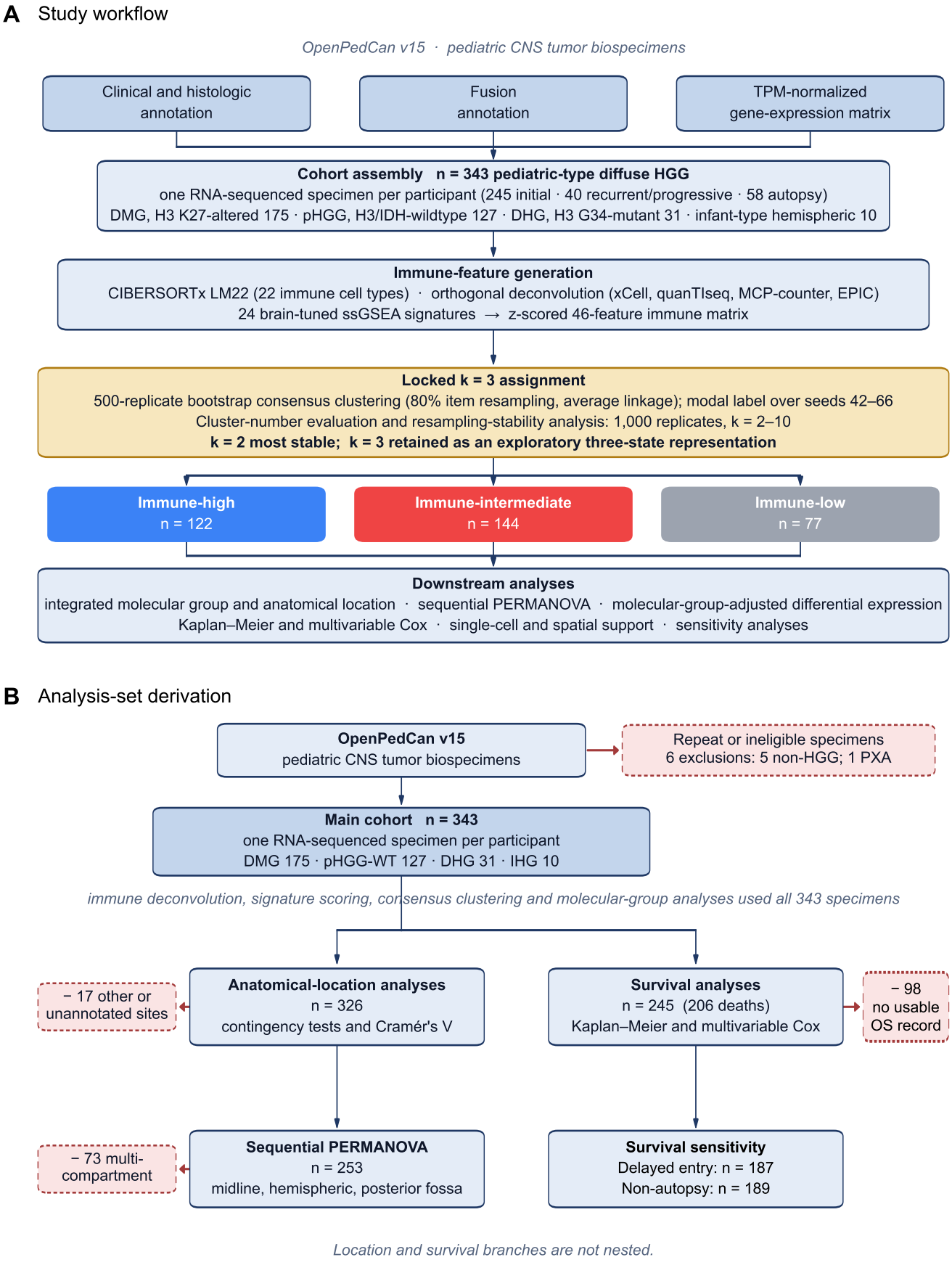

A Study workflow
OpenPedCan v15 · pediatric CNS tumor biospecimens
Clinical and histologic annotation
Fusion annotation
TPM-normalized gene-expression matrix
Cohort assembly n = 343 pediatric-type diffuse HGG
one RNA-sequenced specimen per participant (245 initial · 40 recurrent/progressive · 58 autopsy)
DMG, H3 K27-altered 175 · pHGG, H3/IDH-wildtype 127 · DHG, H3 G34-mutant 31 · infant-type hemispheric 10
Immune-feature generation
CIBERSORTx LM22 (22 immune cell types) · orthogonal deconvolution (xCell, quanTIseq, MCP-counter, EPIC)
24 brain-tuned ssGSEA signatures → z-scored 46-feature immune matrix
Locked k = 3 assignment
500-replicate bootstrap consensus clustering (80% item resampling, average linkage); modal label over seeds 42–66
Cluster-number evaluation and resampling-stability analysis: 1,000 replicates, k = 2–10
k = 2 most stable; k = 3 retained as an exploratory three-state representation
Immune-high
n = 122
Immune-intermediate
n = 144
Immune-low
n = 77
Downstream analyses
integrated molecular group and anatomical location · sequential PERMANOVA · molecular-group-adjusted differential expression
Kaplan–Meier and multivariable Cox · single-cell and spatial support · sensitivity analyses
B Analysis-set derivation
OpenPedCan v15
pediatric CNS tumor biospecimens
Repeat or ineligible specimens
6 exclusions: 5 non-HGG; 1 PXA
Main cohort n = 343
one RNA-sequenced specimen per participant
DMG 175 · pHGG-WT 127 · DHG 31 · IHG 10
immune deconvolution, signature scoring, consensus clustering and molecular-group analyses used all 343 specimens
− 17 other or unannotated sites
Anatomical-location analyses
n = 326
contingency tests and Cramér's V
Survival analyses
n = 245 (206 deaths)
Kaplan–Meier and multivariable Cox
− 98 no usable OS record
− 73 multi-compartment
Sequential PERMANOVA
n = 253
midline, hemispheric, posterior fossa
Survival sensitivity
Delayed entry: n = 187
Non-autopsy: n = 189
Location and survival branches are not nested.